BRAINDANCER · CONTROLS

every button, knob, layer, combo and page

firmware v0.9.1 · M-VAVE FM-1 · norfolkhacker.github.io/braindancer

NAVIGATION · SIX RULES

1 · SELECT WALKS THE PAGES

Every page of the section in one chain, a page a detent, stopping at both ends. No levels, no zoom.

FX: SENDS → SPACE → MASTER → MORE

2 · THE BUTTON: IN, OR ON

From another section: the page you were last on there. Already inside: the next page (round).

SEQ › STEPS: SEQ → STEP EXTRAS · GLO › LIVE: GLO → MIX

3 · PRESETS LOADS PRESETS

Always: the track's sounds; SEQ the patterns (PHRASES, OCT+ loads); GEN the recent pieces. Never a page, a track or a module.

FX › SPACE: PRESETS ▶ the next sound

4 · KNOB 1 PICKS

The step, the slot, the stage, the operator, the kit's sound, the row. The track: always ALGORITHM.

SEQ › STEPS: K1 = all 64 steps · MORE: K1 = the row, K2 its value

5 · HOLD = THE LAYER

Held + SELECT: HOME GEN · EDIT SEQ LFO FX their dice · GLO the tempo · ENV SCL ARP SAVE their chain.

SEQ + G5 CLR STEP · SAVE + C5 NAME

6 · THE SCREEN SAYS WHERE

The header names the section and the track; the status line what the knobs and the next tap do. Footer: the page's place, n / N.

SEQ · STEPS 1 / 4

THE PANEL

KNOBS AND TRANSPORT

MASTER	Master volume.
SELECT	**Walks the section's pages**, a page a detent, stopping at the ends (the MENU: its pages). Held with a button: HOME **GEN**; EDIT · SEQ · LFO · FX their **dice** (OCT+ keeps, OCT– back); GLO the tempo; ENV · SCL · ARP · SAVE their chain.
PRESETS	**Always presets**: the sounds (factory and user) on every page; SEQ: the patterns (PHRASES); GEN: the recent pieces. Nothing in the MENU or with a button held.
ALGORITHM	Selects the track from any page: T1–T6 (T1–T4 with TRACKS 4). MENU: nothing. A layer held: MAP · KNOBS · STATE.
KNOB 1–4	The four knobs on screen, in their colours; KNOB 1 picks (step, slot, row). Fast sweeps the whole range, slow fine-tunes. With a layer held: the layer's. A step held: a lock on that step.
OCT– · OCT+	The keyboard's octave (both together: back to 0); on DRUM the kit stays put: +1/+2/+3 REPEAT 1/2, REPEAT, RATCHET X2; –1 SHORT, –2 GHOST, –3 FLAM. On action rows and dialogs: back / confirm.
PLAY	Start / stop. GLO + PLAY: restart from the top.
REC	Arms live recording on the selected track (notes and knob moves, overdub). On SEQ › STEPS stopped: step recording.
HOME	Tap: HOME (on HOME the next page: TRACKS · VISUAL · GEN). **Hold: the MENU.**
KEYS	27 keys play the selected track; with a layer held they become that layer's; on GLO › LIVE they are the tricks.

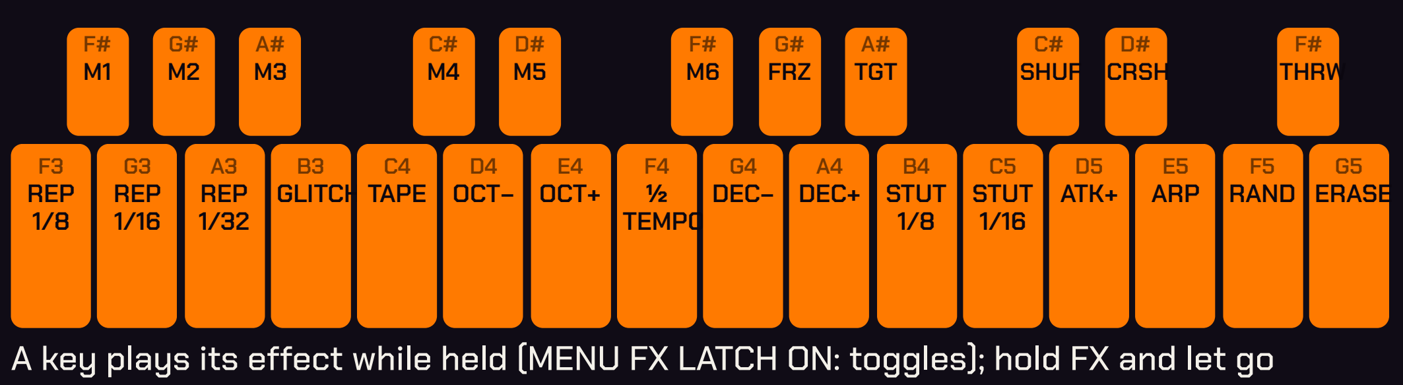
SECTIONS

BUTTON	PAGES (A TAP: THE NEXT · SELECT WALKS THEM)	HOLD
HOME	TRACKS · VISUAL · GEN (ENERGY · FORM · COMPLEX · TAPE; OCT+ GENERATE; PRESETS RECENT)	MENU
EDIT	SOUND · TONE · VOICE or FILTER · MORE (FM6: SOUND · OPERATOR · OP ENV · MORE; DRUM-X: KIT · SOUND · SOUND 2 · MORE)	INIT · RECALL · sound mutes
SEQ	STEPS · STEP EXTRAS · PATTERN · MORE (TRACK · STEP · MUTATE · SHAPE · PLAYHEADS · AUTOMATION)	SEQ TOOLS
FX	SENDS · SPACE · MASTER · MORE (DISTORT · AUTO PAN · DELAY · REVERB · CHORUS · SLICER · SLICE PITCH · PUNCH LANE · MASTER · TAPE)	PUNCH-IN FX
GLO	LIVE · MIX · MIDI · MORE (LIVE: the rails · SYSTEM · TEMPO · MIDI · UNDO); at power-on LIVE	MUTE · SOLO · LEVEL · TAP · UNDO
ENV	AMP · ENV TO	—
LFO	LFO · DEST · MOD MATRIX · MORE (PHASE · STEPS)	MACROS M1–M4 (stay latched)
SCL	KEY · CHORD · MORE	ROOT by key
ARP	ARP · ARP 2	—
SAVE	SOUND · SONG · PROJECT · MORE (from outside, SOUND comes with SAVE picked)	SONG layer · C5 NAME
REC	(transport)	CLEAR · CLICK

BUTTON from elsewhere: the page last shown; inside: the next page. SELECT walks every page of the section. **MORE**: KNOB 1 picks a row, KNOB 2 sets it.

LAYERS · HOLD A BUTTON

FX · PUNCH-IN EFFECTS



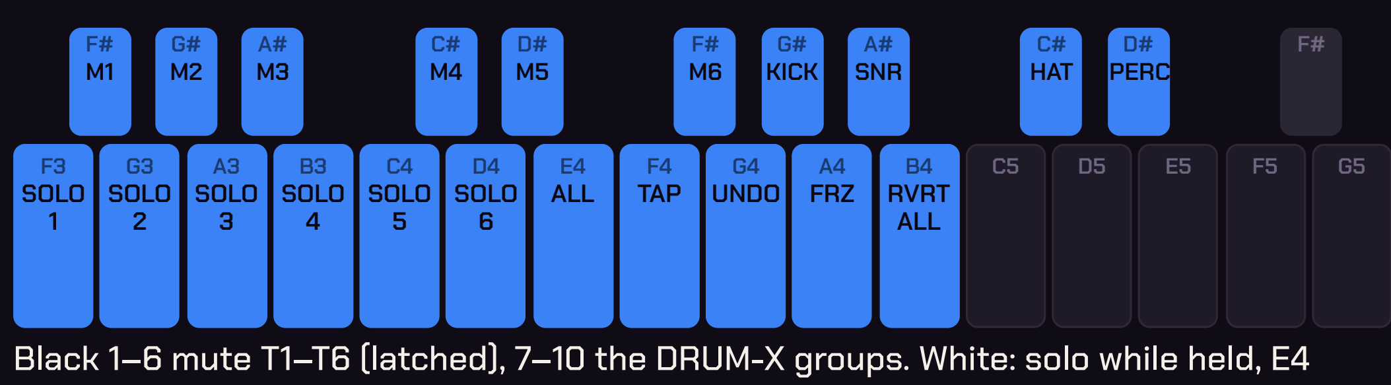
A key plays its effect while held (MENU FX LATCH ON: toggles); hold FX and let go untouched: **LOCKED** open, FX again unlocks; OCT– all off. In on the next 1/16. Black 1–6 mute T1–T6, G#4 FREEZE, A#4 picks which tracks the note effects touch (all · synths · drums), C#5 SHUFFLE; D#5 F#5 sweep CRUSH, THROW dry to wet over 2 bars (back in a beat). With REC armed and playing they record into the section's lane; G5 erases it.

KNOBS · PERFORMANCE MACROS

STEP LOOP	INTERVAL	MUTATE	DJ FILTER

Up to the right; turn any back: all four off at once. GLITCH, TAPE STOP, FREEZE, SHUFFLE held: their own four.

GLO · MIX



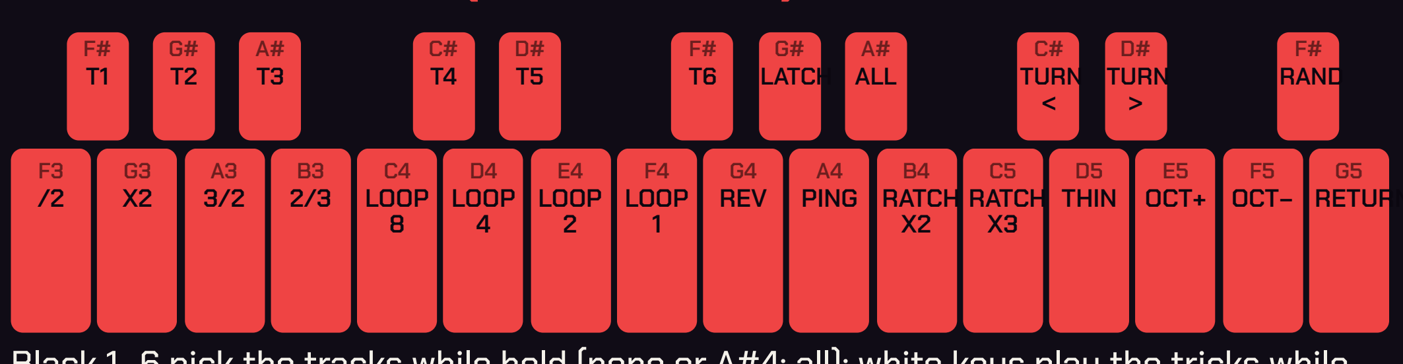
Black 1–6 mute T1–T6 (latched), 7–10 the DRUM-X groups. White: solo while held, E4 unmutes all, F4 tap tempo, G4 **UNDO** the last big change (again: redo), A4 **FREEZE** MUTATE, B4 **REVERT ALL** (every track back to MUTATE's anchor). SELECT: the tempo. PLAY: restart.

KNOBS

LEVEL T1	LEVEL T2	LEVEL T3	LEVEL T4

T5 or T6 selected: the knobs are T3–T6.

GLO › LIVE · TRICKS (ON THE PAGE)

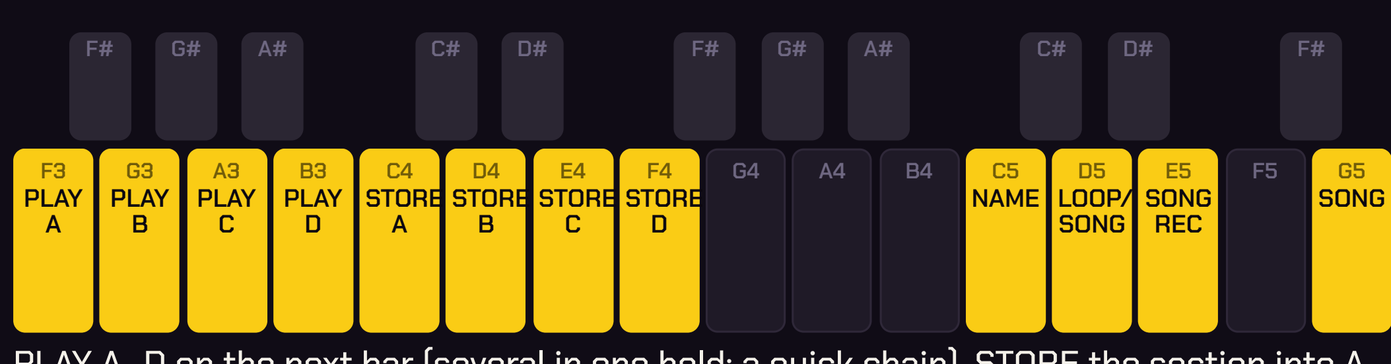


Black 1–6 pick the tracks while held (none or A#4: all); white keys play the tricks while held. G#4 LATCH, C#5 / D#5 TURN, F#5 RANDOM. OCT– tapped RETURN, held BYPASS; OCT+ twice KEEP (into the sequence; GLO + G4 undoes). GLO › MORE › LIVE: the rails (TRACKS, RATE, CUT, DENSITY), latched.

KNOBS

RATE	LOOP	THIN	DJ FILTER

SAVE · SONG

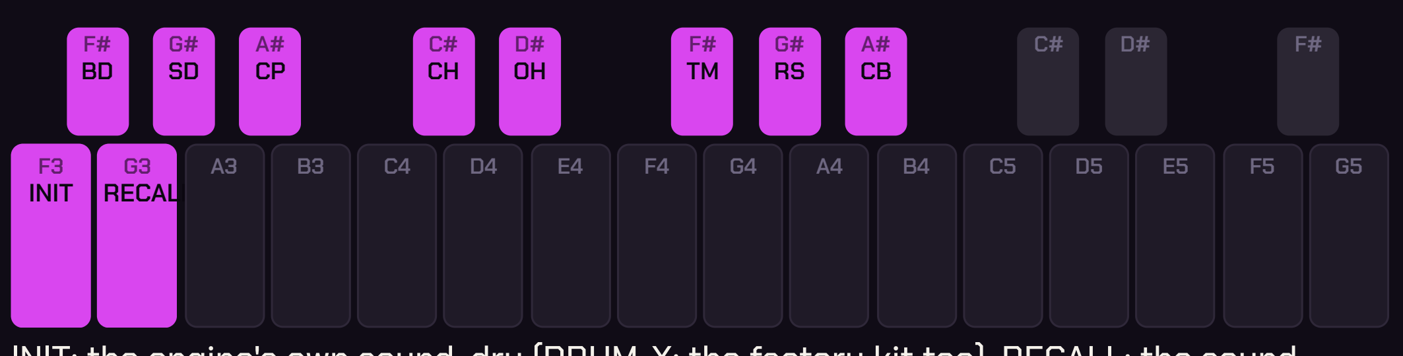


PLAY A–D on the next bar (several in one hold: a quick chain), STORE the section into A–D. OCT+ stores the playing section, OCT– recalls it. C5 renames (SOUND, SONG, PROJECT). On SONG KNOB 4 stores a scene.

KNOBS

PROJECT	-	-	-

EDIT · SOUND



INIT: the engine's own sound, dry (DRUM-X: the factory kit too). RECALL: the sound stored in the section. On a DRUM track black 1–8 mute each DRUM-X sound.

KNOBS

SOUND 1	SOUND 2	SOUND 3	SOUND 4

LFO · MACROS

Knobs M1–M4; they stay latched when let go (LFO again: off). F3 rolls new routes, G3 clears them (only tracks that sound). LFO + SELECT: the track's LFO diced.

KNOBS

M1	M2	M3	M4

SCL · ROOT

Any key sets the ROOT. ROOT and SCALE are every melodic track's.

KNOBS

ROOT	SCALE	ROTATE	PITCH

SEQ · TOOLS

On the SEQ pages: F3 CLEAR · G3 REVERSE · A3 B3 SHIFT · C4 RANDOM · D4 COOK · E4 BEAT · F4 CLR AUTO · G4–C5 the lane's tools (DRUM) · D5 REVERT · G5 CLR STEP · REC: CLEAR ALL. No undo.

KNOBS

LENGTH	DIVISION	SWING	GATE

REC · RECORD

F3 CLEAR the track (at once; GLO + G4 undoes it), G3 the metronome CLICK.

KNOBS

CLICK	-	-	-

COMBOS

BUTTONS TOGETHER

HOME + SELECT	GEN, the song generator (HOME's last pages).
EDIT · SEQ · LFO · FX + SELECT	Their dice: sound · sequence · LFO · effects (BEFORE / NOW; OCT+ keeps, OCT– back).
ENV · SCL · ARP · SAVE + SELECT	Into that section and along its chain.
SEQ + G5	CLR STEP: the picked step and its locks (the AUTOMATION list: deletes the record).
SAVE + C5	NAME: rename the slot, the section, the project.
REC + FX EDIT ENV LFO SCL ARP GLO	Clear that part's automation on the selected track (FX: the punch-in lane too).
SEQ + REC	CLEAR ALL SEQUENCES? (OCT+ confirms).
SAVE + OCT+ / OCT–	Store / recall the playing section.
GLO + PLAY	Restart from the top.
GLO + SELECT	Tempo (from any page; under CLOCK USB or TRS the clock's).
STEP HELD + KNOB	A lock on that step (LOCK · SETTING · VALUE); + SELECT the EDIT page they lock; + EDIT clears its locks.
HOME held	MENU: AUDIO · CLICK · CONTROLS · MORE (speaker EQ, tracks, load, BPM lock; click and count-in; hold time, knob speedup, FX latch, calibration; MORE: screen, LEDs, USB audio, CPU cores and clock, ABOUT).

EVERY PAGE · KNOB 1 – 4

ENV

AMP	ATTACK	DECAY	SUSTAIN	RELEASE
ENV TO	FILTER	PITCH	SHAPE	LOOP

LFO

LFO	RATE	WAVE	MODE	FADE
DEST	PITCH	FILTER	SHAPE	AMP
MOD MATRIX	SLOT	SOURCE	DEST	AMOUNT
MORE	PHASE · STEPS (LENGTH, STEP RATE, SMOOTH, STEP 1–16): K1 picks, K2 sets			

SCL · ARP

KEY	ROOT	SCALE	SNAP	ALL TRACKS
CHORD	CHORD	VOICING	STRUM	-
MORE	TRANSPOSE: K1 picks, K2 sets			
ARP	ARP	RATE	OCTAVES	GATE
ARP 2	SWING	CHANCE	HOLD	ORDER

FX

SENDS	DISTORT	CHORUS	DELAY	REVERB
SPACE	DLY TIME	DLY FDBK	REV TYPE	REV SIZE
MASTER	CLIP	PUNCH	DUCK	AGE
MORE	DISTORT · AUTO PAN · DELAY · REVERB · CHORUS · SLICER · SLICE PITCH · PUNCH LANE · MASTER · TAPE: K1 picks, K2 sets			

EDIT

SOUND	E1	E2	E3	E4
TONE	E5	E6	E7	E8
VOICE	VOICE	GLIDE	PRIORITY	SPREAD
FILTER	FILTER	CUTOFF	RESO	DISTORT
FM6 SOUND	ALGORITHM	FEEDBACK	MOD LEVEL	MOD RATIO
OPERATOR	OP	RATIO	FINE	LEVEL
OP ENV	OP	STAGE	RATE	LEVEL
KIT	MORPH	TUNE	TONE	DECAY
DRUM SOUND	DRUM	LEVEL	PITCH	DECAY
SOUND 2	DRUM	NOISE	COLOR	WAVE
MORE	the engine's other settings in groups: K1 picks, K2 sets			

SEQ

STEPS	STEP	NOTE	ACC/SLD	GATE
STEPS · DRUM	STEP	LANE	HIT	ACC
STEP EXTRAS	RATCHET	CHANCE	CONDITION	MICRO
PATTERN	LENGTH	DIVISION	SWING	GATE
MORE	TRACK (SEQ PITCH, CHANCE, HUMAN, REC SNAP, PLAYS) · STEP (TYPE, TYPE 2, EVERY, FOCUS, PITCH, MOVE) · MUTATE (SOUND, PATTERN, AMOUNT, MODE) · SHAPE (STRETCH, REPEAT, NUDGE, ROTATE) · PLAYHEADS (HEAD, RATE, DIRECTION, INTERVAL) · AUTOMATION (PLAY, EVENTS, LOCKS, CLEAR, the records): K1 picks, K2 sets			

GLO · SAVE · MENU

LIVE	RATE	LOOP	THIN	DJ FILTER
MIX	LEVEL	PAN	SWING	TUNE
MIDI	IN	OUT	CHANNEL	CLOCK
MORE	LIVE (TRACKS, RATE, CUT, DENSITY) · SYSTEM (TRACKS, LOAD, CPU) · TEMPO (BPM) · MIDI (KEYBOARD PLAYS SELECTED, CLOCK OUT, NOTES OUT, LEARN, LEARN CLEAR) · UNDO: K1 picks, K2 sets			
SAVE · SOUND	SLOT	PRESET	NAME	SAVE
SAVE · SONG	SECTION	ROW	BARS	SCENE
SAVE · PROJECT	PROJECT	LOAD	SAVE	NAME
SAVE · MORE	SONG (END, DELETE ROW, CLEAR SONG) · TOOLS (CLEAR SEQ, INIT SOUND, ERASE USER SLOT, NEW PROJECT, CPU TEST) · SOUND (LOAD SLOT, ENGINE, FAVORITE, LIST): K1 picks, K2 sets			
MENU	AUDIO (SPEAKER EQ TRACKS LOAD BPM LOCK) · CLICK · CONTROLS (HOLD TIME KNOB SPEEDUP FX LATCH CALIBRATION) · MORE			

GLO › MIDI

MIDI PER TRACK · THE CLOCK

KNOB 1 **IN**, KNOB 2 **OUT**, KNOB 3 **CHANNEL** (1–16) of the selected track (ALGORITHM picks it; its row lit in the table), KNOB 4 **CLOCK** INT / USB / TRS. A track plays the notes and CCs of its channel while IN is on, and sends its notes on its channel over USB while OUT and NOTES OUT are on. Channel 16's Program Change always picks a section or project. GLO › MORE › MIDI: **KEYBOARD PLAYS SELECTED** (notes on a channel no track listens to play the selected track), **CLOCK OUT** (USB only: the TRS socket is an input), **NOTES OUT** OFF / KEYS / KEYS+SEQ, **LEARN**, LEARN CLEAR. Under CLOCK USB or TRS the tempo is the clock's.

GPL-3.0 · Built on JIANT FM (Juan Jiant) and Felucca (Hügelton Instruments) · song mode from SLOOP